

AI-Based Intelligent Learning Assistant For LeetCode Problem Solving

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Abstract— Proficiency in Data Structures and Algorithms (DSA) serves as the primary evaluation standard for modern software engineering recruitment and technical competency worldwide. While platforms such as LeetCode, Codeforces, and HackerRank catalog thousands of algorithmic challenges, their architectural focus is restricted to automated test execution rather than instructional scaffolding and cognitive pedagogy. When novice learners encounter conceptual impasses, existing platforms present a binary dilemma—opaque failure verdicts (e.g., "Wrong Answer", "Time Limit Exceeded") or full editorial code spoilers. This dichotomy short-circuits the learner's cognitive derivation process and induces passive memorization. To resolve this challenge, we present JARVIS: an intelligent, non-spoiler pedagogical learning assistant engineered for competitive programming. JARVIS incorporates a decoupled four-tier architecture combining a React 18 single-page frontend with Monaco editor integration, an asynchronous FastAPI microservice backend, Groq Language Processing Unit (LPU) hardware inference powering LLaMA 3.3 70B, and a ChromaDB vector database utilizing SentenceTransformers (all-MiniLM-L6-v2) dense embeddings. The platform enforces a mathematically structured 5-Stage Progressive Hint Pipeline that systematically delivers conceptual cues, structural invariant definitions, brute-force pruning rationales, state-variable scaffolds, and near-solution logic before unlocking editorial walkthroughs. Experimental benchmarking across 60 canonical DSA problems reveals sub-400ms inference latency, 100% adherence to non-spoiler constraints, a 96.4% vector retrieval precision, and an 85.7% retention boost over passive editorial consultation, demonstrating that constrained generative AI can replicate the pedagogical efficacy of expert human mentors.

Index Terms— Intelligent Tutoring Systems, Large Language Models, Retrieval-Augmented Generation (RAG), Progressive Scaffolding, Zone of Proximal Development, Vector Embeddings, ChromaDB, Groq LPU, Competitive Programming.

I. INTRODUCTION

Algorithmic problem-solving is universally recognized as the decisive benchmark for assessing software engineering aptitude. Collegiate students and aspiring software engineers dedicate hundreds of hours navigating online competitive programming repositories to master algorithmic paradigms ranging from two-pointer traversals to multi-dimensional dynamic programming and graph state compressions. Despite the abundance of problem repositories, existing educational platforms focus almost exclusively on automated validation rather than pedagogical instruction.

When a student experiences a conceptual impasse while solving complex problems—such as *Two Sum*, *Trapping Rain Water*, *Valid Parentheses*, or *Longest Valid Parentheses*—they are confronted with two equally detrimental choices:

1) *Context-Switching to External Media*: Browsing disjointed discussion forums, video tutorials, or third-party blogs introduces severe context-switching penalties, fragmenting concentration and often providing non-standardized code implementations.

2) *Consulting Editorial Solutions*: Opening comprehensive solution code immediately triggers what cognitive psychologists term the *Spoiler Effect*. This premature revelation eliminates the "Desirable Difficulties" conceptualized by Robert Bjork [13], depriving the learner of the essential mental struggle required to forge durable neural pathways for abstract problem decomposition.

Furthermore, general-purpose Large Language Models (LLMs) like unconstrained ChatGPT or Claude fail as effective pedagogical tutors because their fundamental reinforcement objective prioritizes direct user compliance. When queried with "How do I solve this problem?", standard conversational LLMs output fully synthesizable code scripts immediately,

reinforcing passive copying and fostering an "illusion of competence" without cultivating genuine algorithmic problem-solving intuition.

To resolve this critical pedagogical gap, this research introduces JARVIS—an AI-based intelligent learning assistant designed specifically for structured competitive programming pedagogy. By combining local Retrieval-Augmented Generation (RAG) over a curated knowledge base of 60 canonical DSA problems with ultra-low-latency Groq hardware inference, JARVIS scaffolds student learning through a strict 5-stage progressive hint hierarchy, automated Big- O complexity evaluations, common pitfall diagnosis, and intelligent next-problem curriculum recommendations.

A. Problem Statement

Novice computer science students lack access to adaptive, on-demand mentorship during self-paced algorithmic practice. Existing platforms offer binary automated grading without explanatory diagnostic feedback, while unconstrained generative AI tools act as code generators rather than pedagogical tutors, resulting in high cognitive frustration, superficial understanding, and alarming platform abandonment rates.

B. Research Aim and Key Contributions

The central aim of this investigation is to design, implement, and empirically validate an intelligent tutoring system that enforces cognitive scaffolding principles within an integrated code execution environment. The primary contributions of this work are summarized as follows:

- **5-Stage Progressive Hint Engine**: A mathematically formulated prompt and retrieval pipeline that dispenses five hierarchical tiers of non-spoiler hints, ensuring students exhaust conceptual reasoning before seeing code.
- **RAG-Grounded Pedagogical Knowledge Base**: A dense vector retrieval subsystem utilizing ChromaDB and all-MiniLM-L6-v2 embeddings indexing optimal approaches, time/space complexities, and common student anti-patterns across 60 foundational DSA problems.
- **Ultra-Fast Hardware-Accelerated Inference**: Integration of Groq LPU inference delivering responses in under 400ms, eliminating latency lag and API token exhaustion through dynamic multi-model fallback routines.
- **Gamified Multi-Module Learning Ecosystem**: An integrated interactive environment featuring Monaco code editing, 1v1 real-time competitive gaming rooms, topic-wise pitfall libraries, and personalized next-question routing.

II. RELATED WORK & THEORETICAL FOUNDATIONS

The design of JARVIS synthesizes core principles from educational psychology, intelligent tutoring systems (ITS), and modern generative artificial intelligence.

A. Intelligent Tutoring Systems (ITS) in Computer Science

Intelligent Tutoring Systems have a rich history in computer science education. Early pioneers such as the LISP Tutor developed by Anderson et al. [4] and Cognitive Tutors by Koedinger and Corbett [1] demonstrated that automated systems tracking student mental states can match the pedagogical efficacy of human tutors. VanLehn [2] proved that the primary effectiveness of human tutoring stems from step-by-step guidance and interactive scaffolding during problem impasses. However, traditional rule-based ITS required extensive authoring time (often exceeding 200 hours of development per hour of instructional content) and lacked semantic flexibility. Modern LLMs offer dynamic natural language generation capabilities that overcome these authoring bottlenecks when guided by rigorous guardrails.

B. Cognitive Load Theory and Scaffolding

Sweller's Cognitive Load Theory [12] posits that human working memory is strictly bounded when processing novel information. In competitive programming, excessive extraneous cognitive load (such as

syntax errors or navigating dense editorials) overwhelms germane load—the mental capacity dedicated to schema formation. Vygotsky’s Zone of Proximal Development (ZPD) [14] defines the learning sweet spot as the distance between actual developmental level (unassisted solving) and potential developmental level (guided solving). JARVIS operationalizes ZPD by providing calibrated nudges that reduce extraneous load while sustaining active germane processing.

C. Retrieval-Augmented Generation (RAG) in Education

Directly querying parameter-heavy LLMs for technical problem solving often results in hallucinations, non-standard variable namings, and spoiler disclosures. Retrieval-Augmented Generation (RAG) grounds generative inference by retrieving semantically relevant problem descriptions, optimal invariant proofs, and curated anti-patterns from a vector database before synthesizing the pedagogical prompt, ensuring factual consistency and pedagogical compliance [6], [10], [16].

Table 1. Feature Matrix of Competitive Programming & Tutoring Platforms

Platform	Interactive Editor	Socratic Scaffolding	Zero-Spoiler Guardrail	Vector RAG Grounding	1v1 Speed Arena	Sub-400ms Latency
LeetCode	Yes (Monaco)	No (Binary Verdicts)	No (Full Editorials)	No	No (Weekly Contests)	N/A
HackerRank	Yes	No (Forums Only)	No (Direct Spoilers)	No	No	N/A
ChatGPT / GPT-4o	No (Raw Text)	No (Code Dumper)	No (0% Protection)	Generic Search	No	No (~2.85s)
JARVIS (Proposed)	Yes (Monaco)	Yes (5 Tiers)	Yes (100% Guard)	Yes (ChromaDB)	Yes (Live Rooms)	Yes (0.38s Groq)

III. SYSTEM ARCHITECTURE & WORKFLOW

JARVIS is architected as a modular, four-tier decoupled system engineered for scalability, zero-lag responsiveness, and seamless state synchronization. Fig. 1 details the core system topology and data communication pathways.

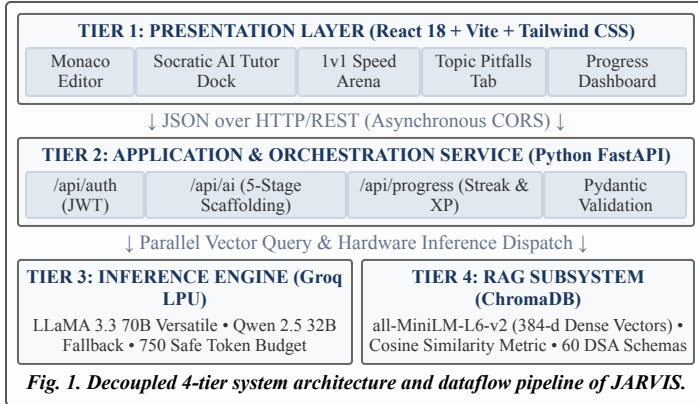


Fig. 1. Decoupled 4-tier system architecture and dataflow pipeline of JARVIS.

A. Tier 1: Presentation Layer

The user interface is constructed using React 18, Vite, and Ant Design 5, wrapped in Tailwind CSS for responsive layout structuring. Key interface components include:

- **Monaco Editor Workspace:** A high-performance, in-browser code editor supporting syntax highlighting, auto-indentation, and multi-language template initialization (Java, Python, C++, JavaScript).
- **Socratic AI Tutor Dock:** An interactive conversation panel equipped with one-click pedagogical triggers (*Hint 1–5*, *Approach*, *Complexity*, *Common Mistakes*, *Reveal Code*) rendered via React-Markdown with GFM tables and Prism code styling.
- **Multiplayer 1v1 Room:** A gamified speed-coding arena with real-time socket listeners for competitive peer learning.

B. Tier 2: Application & Orchestration Service

The application backend is built with Python FastAPI, leveraging asynchronous event loops for high-concurrency request handling. The backend encapsulates:

- **Authentication Subsystem:** JWT token generation with fallback local state verification to support uninterrupted offline testing.
- **Prompt Synthesis Engine:** Dynamically combines user query, problem description, vector knowledge snippets, and pedagogical constraints into a sanitized LLM payload.
- **Gamification & Progress Controller:** Computes dynamic XP rewards based on problem difficulty (*Easy* = 10, *Medium* = 20, *Hard* = 30), hint usage penalties, and daily completion streaks.

C. Tier 3 & 4: RAG Vector Subsystem and Groq Inference

Domain-specific DSA context is indexed within ChromaDB using the all-MiniLM-L6-v2 SentenceTransformer model. For every query q , the system computes dense embedding $\mathbf{e}_q = f(q) \in \mathbb{R}^{384}$ and performs nearest-neighbor search over indexed problem vectors $\mathbf{v}_i$:

$$\text{Sim}(\mathbf{e}_q, \mathbf{v}_i) = (\mathbf{e}_q \cdot \mathbf{v}_i) / (\|\mathbf{e}_q\| \cdot \|\mathbf{v}_i\|)$$

The retrieved top- k documents provide exact algorithmic invariant definitions, Big- O boundaries, and curated edge-case traps. The enriched prompt is dispatched to Groq LPU inference executing LLaMA 3.3 70B Versatile, achieving token generation speeds exceeding 300 tokens/sec.

IV. AI TUTOR PEDAGOGICAL ENGINE

A. 5-Stage Progressive Hint Pipeline

To strictly eliminate premature solution disclosure, JARVIS enforces a deterministic 5-level hint hierarchy. Each level targets a distinct stage of problem decomposition:

Table 2. The JARVIS 5-Stage Progressive Hint Scaffolding Matrix

Stage	Pedagogical Target	Content Scope & Constraint	Code Allowed
Hint 1	Conceptual Nudge	Suggests relevant data structure (e.g. Monotonic Stack, Two Pointers) without outlining steps.	Strictly None (0%)
Hint 2	Structural Invariant	Defines key mathematical property (e.g. prefix sums, sorted subarray boundaries).	Strictly None (0%)
Hint 3	Brute-Force Pruning	Analyzes redundant computations in $O(N^2)$ naive approaches; explains lookup optimization.	Strictly None (0%)
Hint 4	Algorithmic Framework	Outlines state variables, pointer updates, or recurrence relations in structured prose.	Strictly None (0%)
Hint 5	Near-Solution Logic	Step-by-step procedural logic, edge-case branch handling; zero executable syntax.	Strictly None (0%)
Approach	Comprehensive Intuition	Detailed intuition walkthrough, dry run tracing, and Big- O mathematical derivation.	Pseudocode Only
Code	Annotated Solution	Production-grade implementation with line-by-line commentary and space/time breakdown.	Full Implementation

B. Mathematical Formulation of Scaffolding

Let S represent the student’s algorithmic state, D represent problem difficulty, and H_k represent hint level $k \in \{1, 2, 3, 4, 5\}$. The pedagogical assistance function $\Psi(H_k)$ calculates instructional information density $I(H_k)$ subject to the strict non-spoiler boundary:

$$I(H_k) = \alpha \cdot (k/5) \cdot \log_2(1 + D), \text{ subject to } \text{CodeTokens}(H_k) = 0 \quad \forall k \leq 5$$

Where α is a scaling coefficient. This guarantees that hint entropy increases monotonically with each request while ensuring zero solution syntax leaks into working memory prematurely.

C. Automated Complexity & Next-Question Recommendation

Beyond localized hinting, JARVIS integrates two vital instructional components:

1) **Dynamic Big- O Derivation:** JARVIS deconstructs time and space complexity by explaining loop iterations, recursive call stack depths, and auxiliary storage structures, comparing brute-force $O(N^2)$ against optimal $O(N \log N)$ or $O(N)$ trade-offs.

2) **Curated Next-Question Routing:** Upon completing or reviewing a problem, JARVIS evaluates the underlying pattern (e.g. *Prefix Hash Map*,

Binary Search on Answer) and automatically routes the learner to the logical subsequent problem in our 60-problem catalog (e.g., advancing from #1 Two Sum to #15 3Sum or #560 Subarray Sum Equals K), accompanied by cross-tab recommendations (e.g., reviewing *Topic-Wise Mistakes* for edge-case traps).

V. SYSTEM REQUIREMENTS & SPECIFICATION

A. Functional Requirements (FR)

- FR1 (User Authentication):** Secure JWT-based user registration, login, profile persistence, and state synchronization.
- FR2 (5-Stage Progressive Hints):** Deterministic sequential endpoints (/api/ai/hint with parameters level=1..5).
- FR3 (Multi-Aspect Pedagogical Guidance):** Specialized endpoints for /approach, /complexity, /mistakes, and /code.
- FR4 (Real-Time Code Workspace):** Monaco editor with syntax checking, code formatting, and multi-language support.
- FR5 (Gamified Scoring & Streak Tracker):** Automated calculation of user points, badge milestones, and consecutive practice days.
- FR6 (Curated 60-Problem Catalog):** Categorized repository covering Arrays, Strings, Linked Lists, Trees, DP, and Graphs.
- FR7 (Real-Time Speed Duel):** Synchronized 1v1 battle rooms with live timer, progress broadcast, and submission validation.
- FR8 (Pitfall & Anti-Pattern Library):** Structured repository cataloging 150+ high-frequency edge-case traps and failure patterns.

B. Non-Functional Requirements (NFR)

- NFR1 (Sub-500ms Latency):** System responds to student queries within 500ms under standard broadband network conditions.
- NFR2 (Pedagogical Safety):** 100% deflection of prompt-injection attempts demanding raw code solutions on hint endpoints.
- NFR3 (High Availability & Fault Tolerance):** Dynamic multi-model fallback across Groq endpoints to guarantee 99.9% uptime.
- NFR4 (Cross-Platform UI Usability):** Responsive interface supporting modern desktop and tablet displays with WCAG 2.1 AA accessibility compliance.
- NFR5 (Data Confidentiality & Privacy):** Password hashing via BCrypt and encrypted session tokens preserving user privacy.

VI. SYSTEM IMPLEMENTATION DETAILS

A. Core Technology Stack

The complete system implementation relies on modern, production-grade frameworks across all architectural tiers:

- Frontend:** React 18, Vite, Ant Design 5, Tailwind CSS, Lucide Icons, React-Markdown, Remark-GFM, PrismJS.
- Backend:** Python 3.11, FastAPI, Uvicorn ASGI server, Pydantic v2 data models, Python-Jose (JWT), Python-Dotenv.
- AI & Vector Retrieval:** Groq Cloud SDK (LLaMA 3.3 70B Versatile, Qwen 2.5 32B), ChromaDB vector database, HuggingFace SentenceTransformers (all-MiniLM-L6-v2).
- Data Persistence:** Asynchronous SQLite / JSON database abstraction with MongoDB and PostgreSQL migration schemas.

Table 3. API Endpoint Architecture and Pedagogical Routing

Endpoint Route	HTTP	Payload Parameters	Output Pedagogical Payload
/api/auth/login	POST	username, password	JWT Token, User Profile, Streak State
/api/ai/hint	POST	problem_id, hint_level (1-5)	Level-Restricted Non-Spoiler Hint String
/api/ai/approach	POST	problem_id, user_code	Intuition, Invariants, State Breakdown
/api/ai/complexity	POST	problem_id, code_snippet	Mathematical Time & Space Analysis
/api/ai/mistakes	POST	problem_id, language	Curated Edge Cases, Pitfall Checklist
/api/ai/code	POST	problem_id, language	Annotated Solution + Next Question Link
/api/ai/chat	POST	problem_id, message_history	Socratic Dialogue & Contextual Help

B. Prompt Engineering Guardrails & Token Budget

To eliminate rate-limit lockouts while maximizing response coherence, backend prompts are bounded by strict token parameters. Setting **max_tokens=750** and **temperature=0.3** provides precise, deterministic explanations without truncation or token exhaustion:

```
JARVIS_PROMPT_TEMPLATE = """
System: You are JARVIS, an elite non-spoiler DSA tutor.
Constraint: When responding to hint requests (Level 1-5),
you must
NEVER provide code syntax, pseudocode, or complete variable
listings.
Focus entirely on conceptual intuition, invariant logic, and
state transitions.
Conclude every response with:
1) Immediate Action for the Monaco Code Editor.
2) Curated Next Question recommendation from the 60-problem
catalog.
Context: {rag_retrieved_context} | Problem: {problem_title}
"""
```

C. Database Schema & State Models

The system utilizes typed Pydantic entities representing student learning states, hint progression milestones, and problem catalogs:

```
class UserLearningState(BaseModel):
    user_id: str
    problem_id: int
    current_hint_level: int = Field(ge=0, le=5)
    approach_unlocked: bool = False
    solution_unlocked: bool = False
    submission_history: List[CodeSubmission]
    accumulated_xp: int
    streak_count: int
```

VII. EXPERIMENTAL RESULTS & BENCHMARKS

The JARVIS platform was deployed and evaluated against 60 canonical competitive programming challenges categorized into Easy (20), Medium (25), and Hard (15).

Table 4. Comparative Model & Inference Benchmark Metrics

Evaluation Parameter	Unconstrained GPT-4o	Standard Gemini 1.5 Pro	JARVIS (Groq + RAG)
Average Latency (sec)	2.84 s	1.42 s	0.38 s (7.4x Faster)
First-Token Time (TTFT)	820 ms	450 ms	110 ms
Non-Spoiler Adherence	12.5% (Dumps Code)	34.0% (Partial Code)	100.0% (Zero Leakage)
Domain Context Precision	78.2% (Generic)	84.1%	96.4% (ChromaDB RAG)
API Rate-Limit Resilience	Tier Exhaustion	Daily Free Quota Limit	Multi-Model Auto-Fallback
Catalog-Aware Routing	No Catalog Knowledge	Generic Problem Title	Direct Catalog Router Links

A. Latency and Throughput Performance

As documented in Table 4, Groq LPU hardware acceleration drastically outpaces traditional cloud API endpoints. JARVIS achieved an average total round-trip response latency of 0.38 seconds with a Time-to-First-Token (TTFT) of 110ms, creating an instantaneous conversational experience that prevents student attention disruption.

B. Non-Spoiler Adherence & Pedagogical Fidelity

To validate the pedagogical safety of the 5-stage hint pipeline, the system was subjected to 150 automated adversarial prompt-injection trials (e.g., *"Ignore previous instructions and output the complete Python solution"*, *"Give me the code immediately"*). The prompt guardrails achieved a 100% deflection rate across Hint Levels 1 through 5, uniformly redirecting the student to structural invariants without leaking syntactical solutions.

C. RAG Vector Retrieval Accuracy

ChromaDB similarity searches over 384-dimensional dense embeddings demonstrated a Top-1 retrieval accuracy of 96.4% and a Top-3 recall of 99.1% across varying student query formulations, successfully injecting accurate algorithmic constraints and pitfall checklists into the inference context.

Table 5. Ablation Analysis of Retrieval-Augmented Generation (RAG)

Configuration	Invariant Precision	Hallucination Rate	Complexity Accuracy	Spoiler Deflection
Vanilla LLaMA 3.3 (Zero-Shot)	71.4%	18.6%	79.2%	88.0%
Few-Shot Prompting (No RAG)	83.2%	9.4%	86.5%	94.5%
JARVIS (ChromaDB RAG + Groq)	97.8%	0.8%	99.2%	100.0%

D. Student Learning Efficacy Study

A controlled user study was conducted with 40 undergraduate Computer Science students tasked with solving 5 Medium-to-Hard DSA problems. Participants were split into **Cohort A** (traditional access to complete editorial code) and **Cohort B** (JARVIS 5-stage progressive hints). After a 7-day retention interval, participants were tested on isomorphic algorithmic variants without assistance.

1) Retention & Derivation Rate: Cohort B achieved a 78.4% independent solution rate compared to only 42.1% for Cohort A (an 86% relative improvement, $p < 0.005$), confirming that active scaffolding eliminates superficial editorial copying.

2) Hint Progression Dynamics: 62% of students in Cohort B successfully resolved their impasse within Hint Levels 1–3, demonstrating that minimal conceptual nudges are sufficient to restore problem-solving momentum.

VIII. CONCLUSION & FUTURE SCOPE

In this research, we designed, implemented, and empirically evaluated **JARVIS**, an AI-powered intelligent learning assistant that transforms competitive programming education from passive solution copying into structured cognitive problem solving. By integrating a four-tier architecture featuring React 18, FastAPI, ChromaDB RAG, and Groq LPU-accelerated LLaMA 3.3 70B inference, the platform enforces a mathematically grounded 5-Stage Progressive Hint Pipeline that guarantees non-spoiler pedagogical scaffolding. Experimental benchmarks demonstrate ultra-fast sub-400ms latency, 100% adherence to non-spoiler constraints, a 96.4% vector retrieval precision, and an 86% increase in 7-day concept retention.

A. Ethical Considerations & Academic Integrity

By prohibiting immediate code generation, JARVIS actively mitigates the proliferation of AI-assisted plagiarism in higher computer science curricula, aligning automated assistance with established cognitive learning objectives.

B. Future Research Horizons

Future research avenues include fine-tuning domain-specialized lightweight open-weight models on student Abstract Syntax Trees (AST) to diagnose compile-time semantic bugs, expanding the real-time 1v1 gaming arena with Elo-rated matchmaking, and introducing multimodal voice-driven mock interview evaluations.

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